



Cylinder Distribution Platform

Backend API

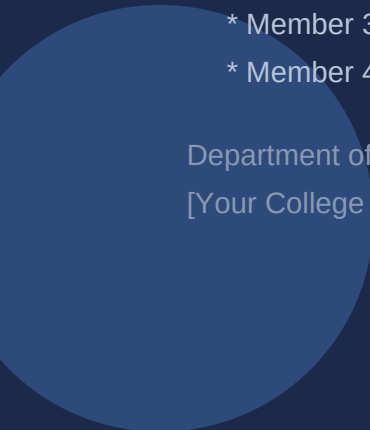
Production-Grade LPG Delivery Service Platform

Team CylDist

Team Members:

- * Member 1 - Full Stack Developer
- * Member 2 - Backend Engineer
- * Member 3 - DevOps & Database
- * Member 4 - Frontend & Testing

Department of Computer Engineering
[Your College Name], 2025



Introduction & Objectives

Overview of the LPG Cylinder Distribution Platform

Platform Overview

- * Digital LPG cylinder booking & delivery management system
- * Multi-role platform:
Admin, Customer, Delivery Agent
- * Real-time GPS tracking & live chat
- * End-to-end order lifecycle management
- * WhatsApp-based booking support
- * Emergency crisis prioritization

Key Objectives

- * Digitize traditional LPG distribution
- * Provide real-time GPS delivery tracking
- * Enable emergency-aware allocation
- * Build secure & scalable architecture
- * Production-ready Docker/K8s deploy
- * Implement role-based access control
- * Create comprehensive API documentation
- * Enable real-time communication

Key Highlights

Node.js + Express.js | MongoDB + Redis | Socket.IO Real-Time | Docker & Kubernetes
JWT Auth + RBAC | Swagger API Docs | WhatsApp Integration | Crisis Engine

Problem Statement & Proposed Solution

Problems in Traditional System

- X Manual booking via phone/visits
- X No real-time delivery tracking
- X Inventory management is opaque
- X No priority during emergencies
- X Paper-based record keeping
- X No communication channel
- X No analytics or reporting
- X Inefficient route planning



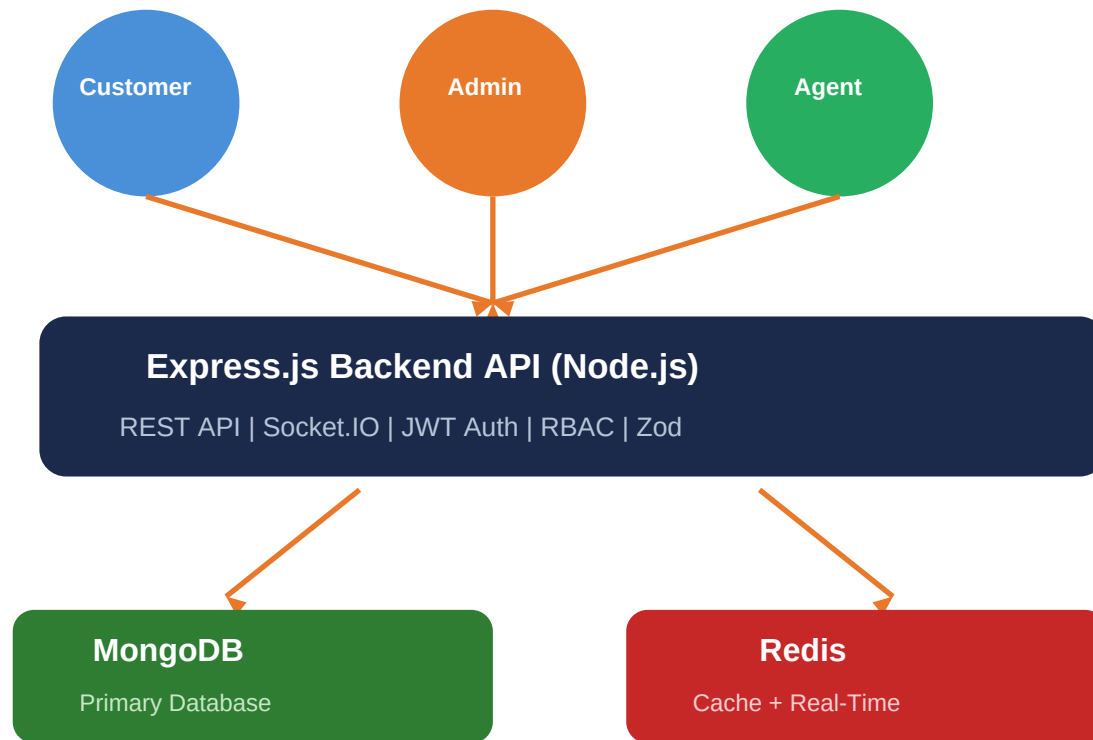
Our Digital Solution

- + Online booking + WhatsApp API
- + Live GPS delivery tracking
- + Automated inventory system
- + Crisis prioritization engine
- + Digital order management
- + Real-time agent-customer chat
- + Dashboard analytics & reports
- + Smart warehouse allocation

Impact: 100% digital workflow | Real-time visibility | Emergency-aware | Scalable microservices

System Architecture & Workflow

User -> Backend API -> Database



Order Lifecycle Flow

1. Customer places order
2. Admin assigns agent
3. Agent picks up cylinder
4. GPS tracking begins
5. Live chat enabled
6. Delivery + OTP verify
7. **Order completed!**

Technologies Used

Modern Tech Stack for Production-Grade Platform

Node.js

Runtime Engine

Express.js

Web Framework

MongoDB

NoSQL Database

Redis

Cache & RT Store

Socket.IO

WebSocket Layer

Docker

Containerization

Kubernetes

Orchestration

Supporting Technologies & Libraries

JWT | bcrypt | Zod | Winston | Swagger | Helmet | CORS | Razorpay | Twilio | SendGrid

AWS S3 | Jest | Supertest | Nodemon | Docker Compose | HPA | Multer | Morgan

Unique Features Implemented

10 Key Feature Highlights

WhatsApp Booking - Book via API

Product Store - Buy accessories

GPS Tracking - Real-time location

Live Chat - Agent messaging

Crisis Engine - Emergency priority

Inventory Mgmt - Warehouse tracking

JWT + RBAC - Role-based security

Notifications - SMS/Email/Push

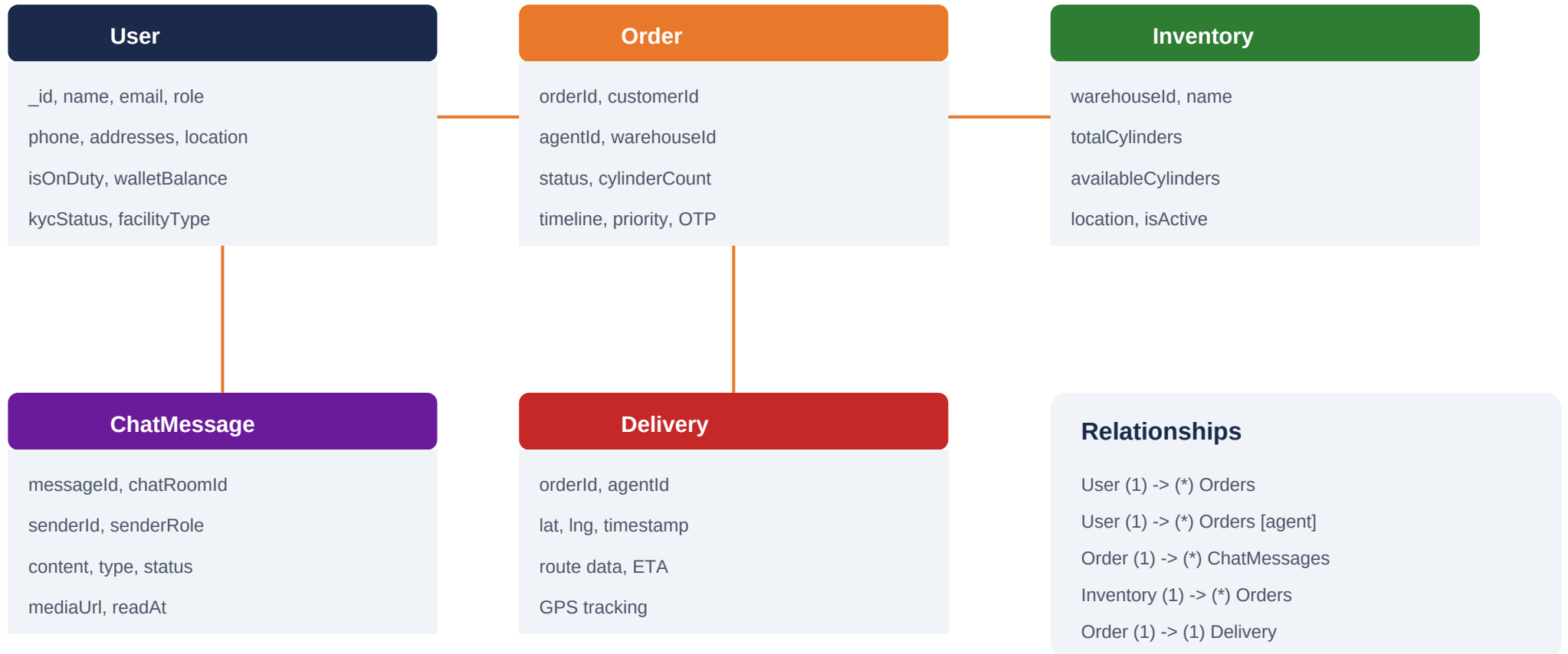
Swagger Docs - Interactive API

Docker + K8s - Container deploy

All features are production-ready with comprehensive error handling & logging

Database Design & ER Diagram

MongoDB Collections & Relationships



Demo & Security Features

API Documentation, Tracking & Security Measures

Live Demo Highlights

Admin Dashboard

- * Order management & analytics
- * Agent assignment & monitoring

Swagger API Documentation

- * /api/v1/docs - Interactive testing
- * All 30+ endpoints documented

Real-Time Delivery Tracking

- * GPS updates every 5 seconds
- * Socket.IO room-per-order

Security Architecture

JWT Authentication

Access (15min) + Refresh (7d)

Rate Limiting

100 req/15min, 10/15min auth

Password Encryption

bcrypt 12 rounds hashing

Role-Based Access (RBAC)

Admin | Customer | Agent

Additional Security Measures

Helmet (CSP) | CORS Allowlist | HPP Protection | Mongo Sanitize | XSS Clean | 10kb Payload Limit

Refresh Token Rotation | Reuse Detection | NoSQL Injection Prevention | Winston Logging | ELK-Ready

Conclusion & Future Scope

Project Achievements

- + Production-grade REST API
- + Real-time GPS + Live Chat
- + Crisis Prioritization Engine
- + Complete Order Lifecycle
- + Docker & K8s Deployment
- + Comprehensive Security
- + WhatsApp Integration
- + Swagger API Documentation
- + Multi-role RBAC System
- + Inventory Management

Future Enhancements

AI Demand Prediction

ML-based consumption forecast

Mobile App Integration

React Native / Flutter app

Online Payment Gateway

Razorpay/Stripe integration

Smart Route Optimization

Google Maps route planning

IoT Cylinder Monitoring

Smart sensors for gas level

Thank You!

Questions & Feedback Welcome